

# Structural and Diffusion MRI Preprocessing Pipeline for the Parkinson's Progression Markers Initiative (PPMI) Dataset

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## Motivation

This guide describes an end-to-end preprocessing pipeline for downstream connectomic analysis of structural MRI and diffusion-weighted MRI (dMRI) images from the Parkinson's Progression Markers Initiative (PPMI) longitudinal MRI dataset, provided by the Michael J. Fox Foundation. The pipeline integrates structural MRI and dMRI processing for connectivity analysis using defined regions of interest (ROIs). Automatic segmentations from structural images were registered to dMRI, where connectivity between regions can be measured using tractography.

Structural T1-weighted images were processed using FreeSurfer's [recon-all](#) pipeline for cortical and subcortical segmentation. These segmentations were matched and propagated to preprocessed dMRI using transforms from FreeSurfer's dMRI reconstruction pipeline, [dt\\_recon](#). Depending on the available data, susceptibility distortions in dMRI were corrected using either FSL's [topup](#) or FreeSurfer's [mri\\_synthmorph](#). A flowchart, outlining the different processing paths and the decision tree used, is included as a supplemental figure (at the end of the report).

## Summary of tools used

| Tool           | Version            | Runtime       |
|----------------|--------------------|---------------|
| dcmunpack      | FreeSurfer v.7.4.1 | 10-15 minutes |
| recon-all      | FreeSurfer v.7.4.1 | 4-6 Hours     |
| topup          | FSL v.6.0.4        | 40-50 minutes |
| mri_synthmorph | FreeSurfer v.8.0.0 | 5-15 minutes  |
| dwidenoise     | MRtrix3 v3.0.4     | 5-10 minutes  |
| mrdegibbs      | MRtrix3 v3.0.4     | ~5 minutes    |
| dt_recon       | FreeSurfer v.7.4.1 | 20-30 minutes |

These times were based on running the analysis on the CPU with 4 cores and 32GB of memory.

## Unpacking DICOM Data

Raw DICOM MRI acquisitions, tagged as T1-weighted or dMRI were downloaded from Laboratory of Neuro Imaging (LONI) Image & Data Archive, [ida.loni.usc.edu](#). DICOM files were subsequently batch-converted to NIfTI format using FreeSurfer's [dcmunpack](#) (v.7.4.1).

```
dcmunpack -scanonly <log file path> -src <DICOM_path> -targ <DICOM path> -  
auto-runseq nii.gz -no-exit-on-error -dcm2niix-createBIDS
```

In addition to the NIfTI volumes, conversion of diffusion-weighted images generated three auxiliary files: two containing the diffusion gradient information (b-values and b-vectors), and one JSON file

with metadata from the DICOM header. The metadata provided information about the phase-encoding direction and total readout time of the sequence, both of which are required for susceptibility distortion correction using FSL's [topup](#).

## Automatic Segmentation with FreeSurfer's `recon-all`

All structural T1-weighted images were processed using FreeSurfer's [recon-all](#) pipeline (v.7.4.1), to generate surface reconstructions and cortical and subcortical segmentations.

```
recon-all -all -i <input T1> -s <subject ID> -sd <subjects directory>
```

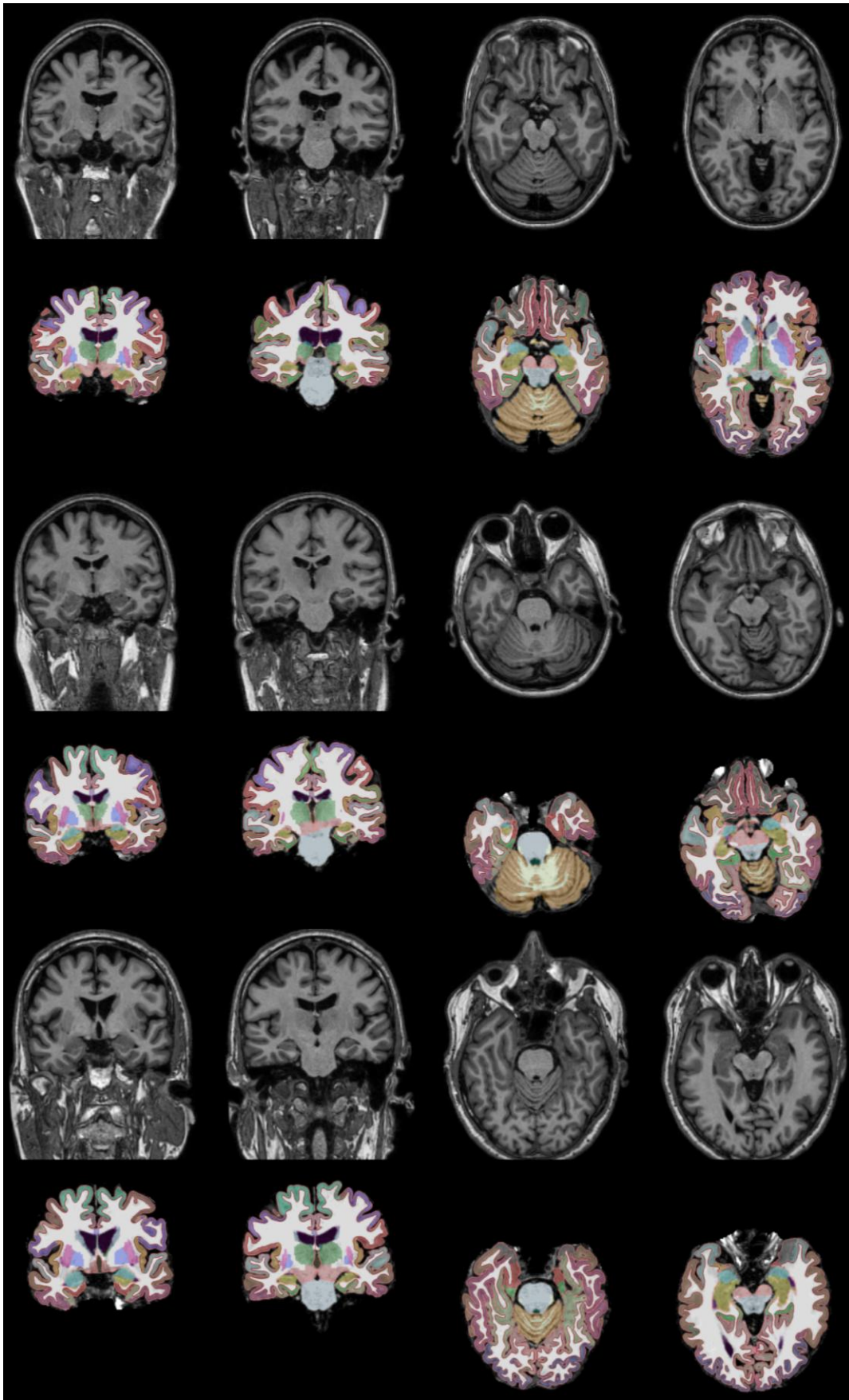
Additionally, `recon-all` served as an initial quality control (QC) step. Structural images that failed processing were identified by parsing the `recon-all-status.log` files and excluded from further analysis. We present in Figure 1 three examples of successful reconstructions from structural images, including the input T1-weighted image alongside its segmentation and cortical surfaces. To avoid selection bias, the images from the first three available subjects in the dataset are shown.

Segmentation quality was manually reviewed to further curate the data. To accelerate the manual QC process, FreeSurfer's `freeview` was used to automatically generate screenshots for each input T1-weighted image and its corresponding segmentation. Screenshot acquisition was done in a headless environment using [fsxvfb freeview](#), reducing processing time by avoiding GUI overhead.

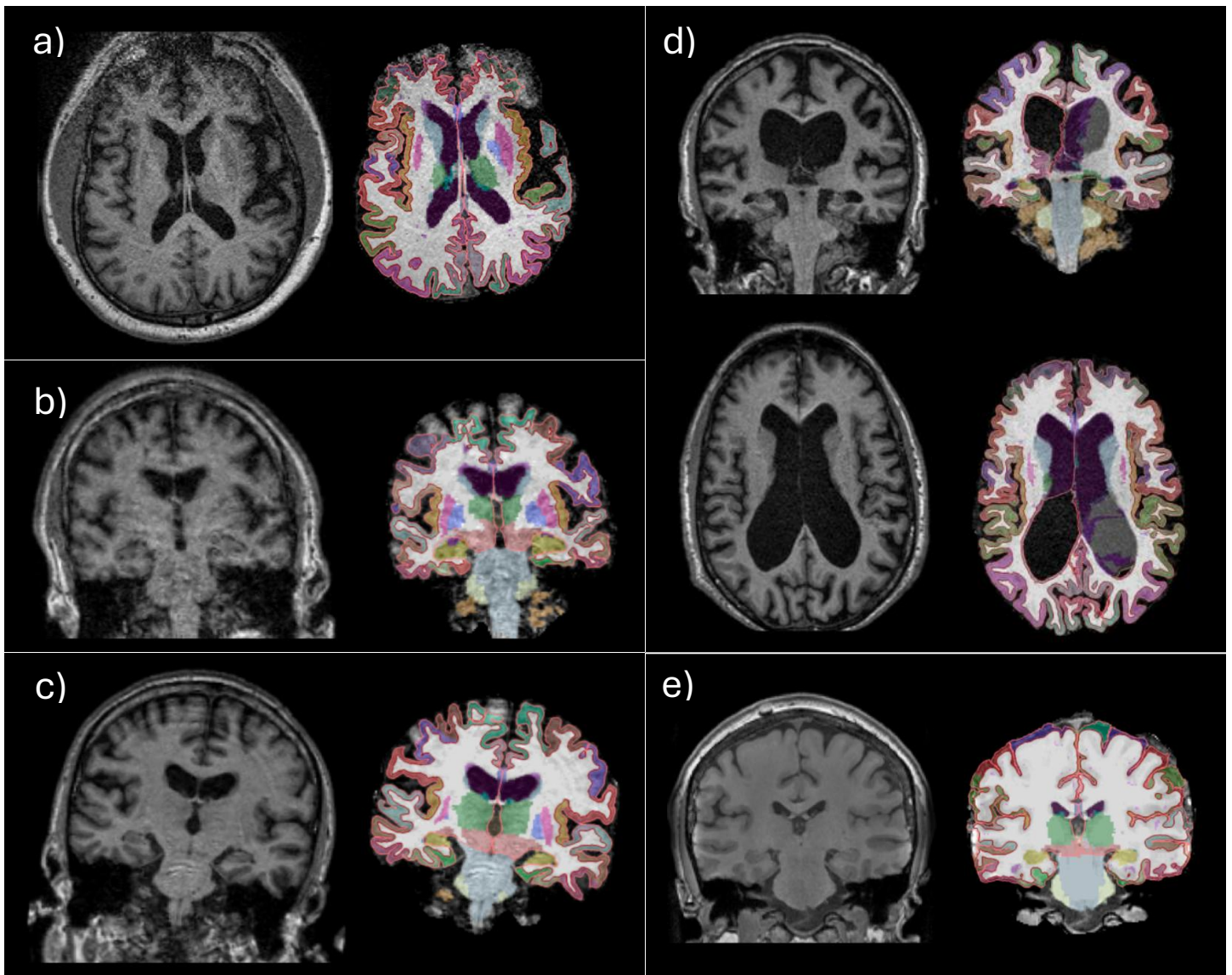
FreeSurfer's `freeview` supports acquiring screenshots of image volumes using the `-ss` flag, and using other flags like `-slice` and `-viewport`, we can easily automate screenshot acquisition. Here we include an example command to generate a coronal screenshot.

```
fsxvfb freeview -v preprocessed_dmri.nii.gz \  
aseg_moved.nii.gz:colormap=lut:opacity=1.0 \  
--layout 1 -zoom 1 --viewsize 1024 1024 \  
--viewport coronal -slice 64 40 40 \  
-ss ${output_directory}/qc_screenshot.png 1 autotrim
```

Image artifacts such as motion, signal noise, or weak tissue contrast can degrade the performance of the automatic segmentation. Figure 2 shows examples of images that failed QC because of inaccurate segmentations.



**Figure 1.** Each row shows an example pair of a structural MRI scan and its corresponding segmentations and cortical surfaces from FreeSurfer's *recon-all*.



**Figure 2.** Examples of failed segmentations from `recon-all`. a-b) segmentation is missing regions of anatomy due to noise, c) missing anatomy from high contrast motion artifacts, d) failed segmentation due to enlarged ventricles, and e) errors in segmentation from low gray-white matter contrast.

## Susceptibility Correction

Magnetic susceptibility artifacts from B0 field inhomogeneities are a significant source of geometric distortion in dMRI, particularly at the inferior frontal cortex and temporal lobes. These distortions are nonlinear and dependent on the phase encoding direction during acquisition. Because `dt_recon` performs a rigid registration of structural and diffusion images, these nonlinear susceptibility distortions must be corrected to ensure accurate correspondence between the dMRI volumes and the segmentations on the structural MRI.

### *Computing Susceptibility Field Using FSL's `topup` (v.6.0.4)*

The most common tool for susceptibility distortion correction is FSL's `topup`, which can correct distortions when two images with opposing phase encoding directions are available. In addition to correcting for spatial distortions, `topup` also performs intensity modulation to account for signal pooling which occurs in areas of distortion. Reverse phase-encoding images were matched based on their acquisition date. Information on the phase-encoding direction of the images was found in the metadata JSON file obtained from the NIfTI conversion steps.

Although both images with opposite phase-encoding directions typically contain the same number of diffusion frames with identical b-values, some sessions contained only b=0 images, presumably to speed up acquisition time. This does not affect the computation of the susceptibility field using `topup`, but can lead to a mild decrease in the signal-to-noise ratio of the final corrected image (since the diffusion-weighted images will not be averaged across the two sets).

To prepare the inputs for `topup`, FreeSurfer's `mri_convert` and `mri_concat` were used to generate an image with only the b=0 frames of both (opposite phase-encoding) images. We used strictly b=0 images even though low-b images (e.g., b<50) are also commonly used with `topup`. To calculate the susceptibility field using `topup`, we created a parameter file, `params.txt`, which includes the phase-encoding information and the total readout time obtained from the metadata. The file is a matrix where each row corresponds to a single b=0 volume in our input image, whose first three columns correspond to the phase encode direction (1 0 0 for left-right, -1 0 0 for right-left, 0 1 0 for anterior-posterior, and so on), and the final column is the total readout time in seconds.

```
mri_convert <input image> <b0 image> -f <indices of b0 frames>
mri_concat --i <b0 image1> --i <b0 image2> --o <topup input>
topup -config=b02b0.cnf --imain <topup input> --datain=acqparams.txt --out
<output> --iout=<corrected b0> --fout=<susceptibility correction field>
```

FSL's `topup` required images with even dimensions on all axes. Therefore, dMRI volumes with odd dimensions were cropped prior to processing. Using the `slicer` attribute from the Nibabel Python library, dMRI volumes were cropped when necessary. Nibabel's `slicer` attribute allows for easy image cropping while keeping images in the same coordinate system. Below, we show a code snippet that crops dMRI volumes with uneven axis dimensions.

```
def cropForTopup(niftiPath):
    niftiObj = nib.load(niftiPath)
    niftiDims = niftiObj.header.get_data_shape()
    xEnd, yEnd, zEnd = niftiDims[:3]

    if xEnd % 2 == 0 and yEnd % 2 == 0 and zEnd % 2 == 0:
        return niftiPath
    else:
        xEnd = xEnd - 1 if xEnd % 2 != 0 else xEnd
        yEnd = yEnd - 1 if yEnd % 2 != 0 else yEnd
        zEnd = zEnd - 1 if zEnd % 2 != 0 else zEnd
        croppedNifti = niftiObj.slicer[0:xEnd, 0:yEnd, 0:zEnd, :]
        newName = niftiPath.split('.nii.gz')[0] + '_cropped.nii.gz'
        nib.save(croppedNifti, newName)
```

### ***Preprocessing Correcting for Noise and Gibbs Ringing***

Prior to applying the susceptibility field from `topup`, the original dMRI volumes were denoised using [dwidenoise](#) and corrected for Gibbs ringing artifacts using [mrdegibbs](#) from the MRtrix3 (v3.0.4) software package. These steps reduce noise and ringing artifacts, improving data quality for any downstream task.

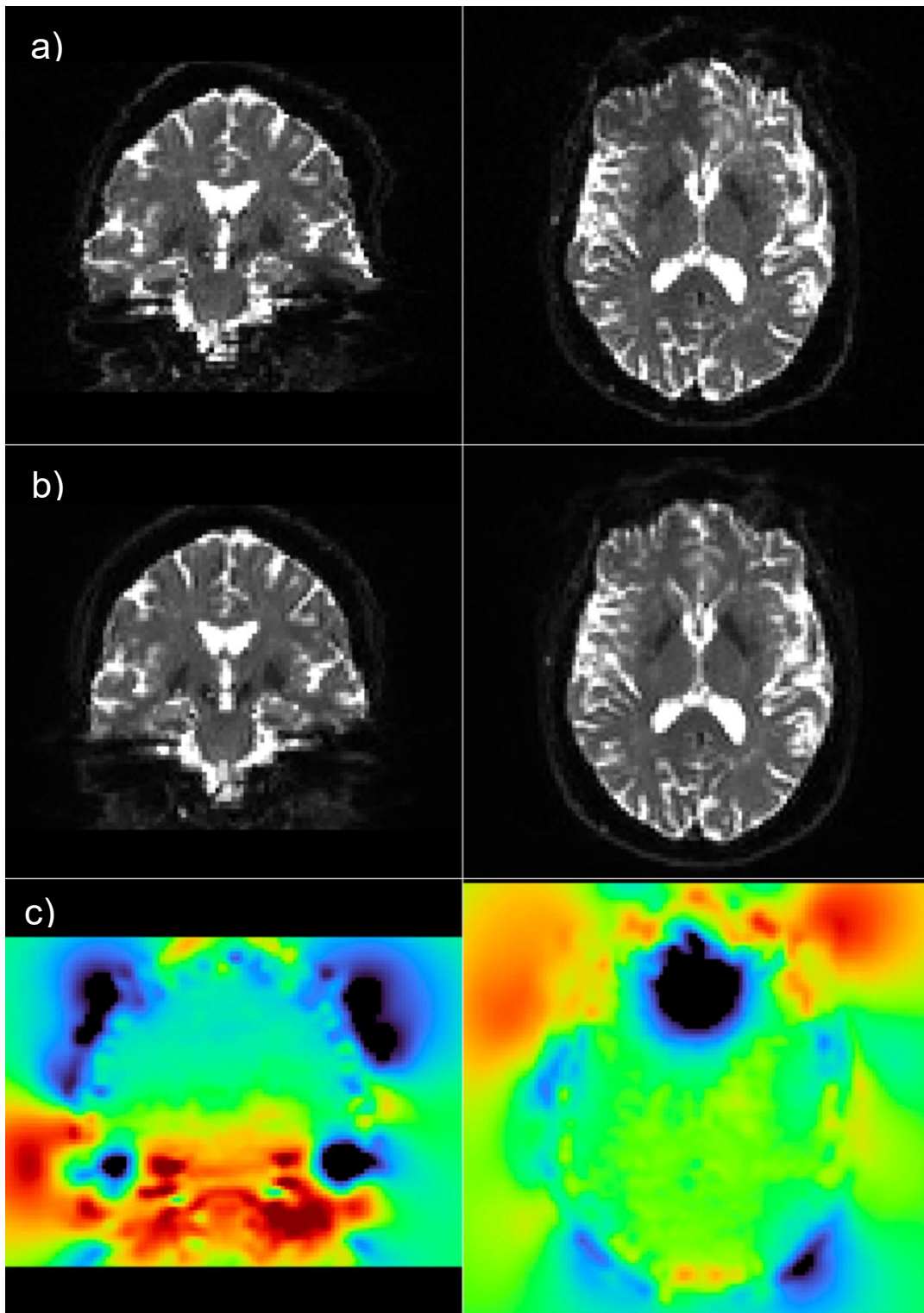
```
dwidenoise -noise dwi_noise.mif dwi.nii.gz dwi_clean.nii.gz
mrdegibbs -axes <slice axes> dwi_clean.nii.gz dwi_degibb.nii.gz
```

### ***Applying the topup deformation field***

After denoising and Gibbs correction, the susceptibility field estimated by `topup` was applied to the dMRI volumes. The following example shows a case where we had dMRI frames for both opposite phase-encoding directions.

```
applytopup --imain=dwi_pos_degibb.nii.gz,dwi_neg_degibb.nii.gz --
datain=acqparams.txt --topup=topup_results --inindex=1,4 --method=jac --
interp=spline --out=corrected_DWI
```

Next, we show in Figure 3 a before/after example of susceptibility correction using `topup` alongside the resulting deformation field.



**Figure 3.** Example results from susceptibility corrections using FSL `topup`. a) original  $b=0$  volumes. b) `topup` corrected image. c) computed deformation field.

### **FreeSurfer's SynthMorph (v.8.0.0)**

When reverse phase encoding acquisitions were not available, susceptibility distortion correction was done using [mri\\_synthmorph](#), following processing with FreeSurfer's dMRI reconstruction pipeline, `dt_recon`. SynthMorph is a deep-learning-based, contrast-agnostic registration tool capable of registering brain MRI images across contrasts. This tool estimates a nonlinear deformation between the T1-weighted and diffusion images, which can be used to correct for susceptibility distortions. Prior to SynthMorph, the `dt_recon` pipeline corrected for eddy currents and motion, and generated a rigid registration of the structural T1-weighted image to diffusion space (see the next Section for details). Using the rigid transform (`register.dat`) generated by `dt_recon`, the intensity normalized T1-weighted image (`nu.mgz`) and the automatic segmentation (`aprac+aseg.mgz`) were resampled to the diffusion space and resolution.

```
mri_vol2vol --mov lowb.nii.gz --targ <structural recon-  
all>/mri/aparc+aseg.mgz --inv --interp nearest --o aparc+aseg.nii.gz --reg  
register.dat --no-save-reg  
  
mri_vol2vol --mov lowb.nii.gz --targ <structural recon-all>/mri/nu.mgz --inv  
--o nu.moved.nii.gz -reg register.dat --no-save-reg
```

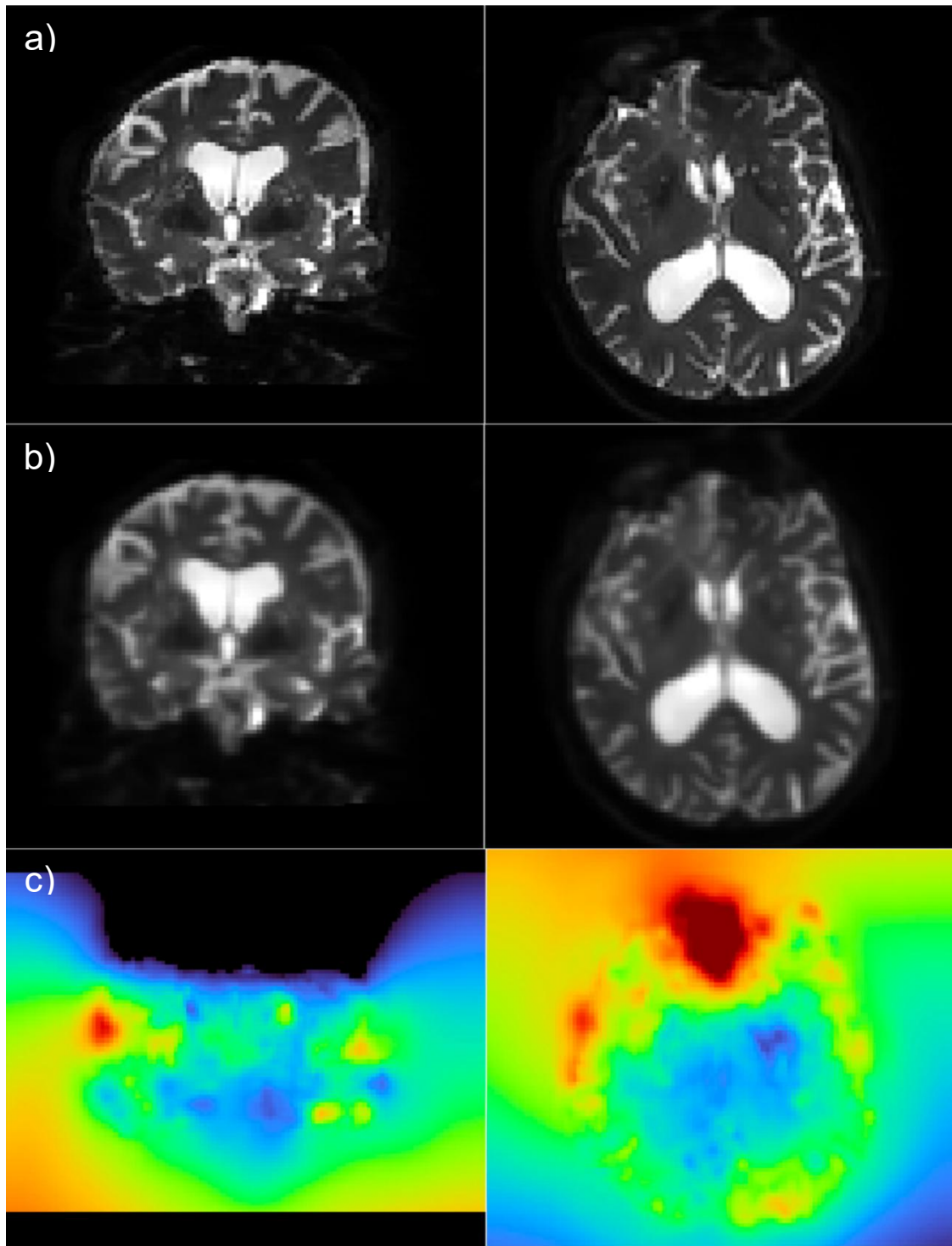
The nonlinear deformation was then computed on the average of all  $b=0$  frames to improve signal-to-noise ratio and minimize errors caused by lingering motion between frames after `dt_recon`. All indices of the  $b=0$  frames were identified using the b-value metadata, extracted with `mri_convert`, and subsequently averaged using `mri_average`.

```
mri_convert dwi-ec.nii.gz dwi-ec.flt.nii.gz -odt float  
mri_convert dwi-ec.flt.nii.gz dwi-ec.b0.<frame index>.nii.gz -f <frame index>  
mri_average -noconform <All b0 files separated by a space> dwi-ec.b0.nii.gz
```

Finally, SynthMorph was used to compute a nonlinear deformation between the average  $b=0$  volume and the resampled structural image. The nonlinear deformation was then applied to the full dMRI volume to correct for susceptibility artifacts.

```
mri_synthmorph register -t dwi-to-struct.synthmorph.nii.gz dwi-ec.b0.nii.gz  
<structural recon-all path>/mri/nu.moved.nii.gz  
  
mri_synthmorph apply dwi-struct.synthmorph.nii.gz dwi-ec.flt.nii.gz dwi-  
ec.corrected.synthmorph.nii.gz
```

Examples of the results are shown in Figure 4.



**Figure 4.** Example results from susceptibility corrections using SynthMorph. a)  $b=0$  from original diffusion image. b) SynthMorph corrected image. c) Deformation field in the left-right (phase-encoding) axis.

The largest deformations were observed in areas with the most pronounced susceptibility artifacts. Without any correction, these distortions would lead to misalignment between the dMRI image and structural segmentation, resulting in mislabeled anatomy. Although SynthMorph effectively reduces these distortions, the nonlinear deformation introduces some spatial smoothing to the corrected images. Alternate approaches for susceptibility correction were also evaluated, including generating synthetic T1-weighted images before doing classical nonlinear registration. However, the SynthMorph approach produced better qualitative results.

The SynthMorph approach corrects geometric distortions but does not perform the intensity modulation (resulting from signal pooling in distorted areas) that `topup` does. This, however, is not expected to significantly affect the dMRI analysis, given that the ratio between the diffusion-weighted and  $b=0$  images ( $S/S_0$ ) is used in subsequent diffusion tensor or orientation distribution function reconstruction, and modulation of both images by a similar value would not affect the ratio.

### FreeSurfer's `dt_recon` (v.7.4.1)

Following denoising and Gibbs ringing correction, `dt_recon` was used to correct for eddy currents (through FSL's `eddy_correct`) and motion across dMRI frames. Additionally, `dt_recon` generates rigid registration results from structural to diffusion space, which we use to resample the automatic segmentation onto the dMRI.

```
dt_recon --i dwi.nii.gz --b <bvals> <bvecs> --s <subject> --o <out directory>

mri_vol2vol --mov lowb.nii.gz --targ <structural recon-
all>/mri/aparc+aseg.mgz --o aparc+aseg.nii.gz -reg register.dat --no-save-reg
--inv
```

The final corrected dMRI volumes were bias-corrected using MRtrix3's `dwibiascorrect` to correct for B1 bias and normalize intensities across the brain.

```
dwibiascorrect ants dwi-ec.corrected.nii.gz dwi-ec_biascorrected.nii.gz -bias
dwi-ec_biasestimate.nii.gz -fslgrad bvecs.dat bvals.dat
```

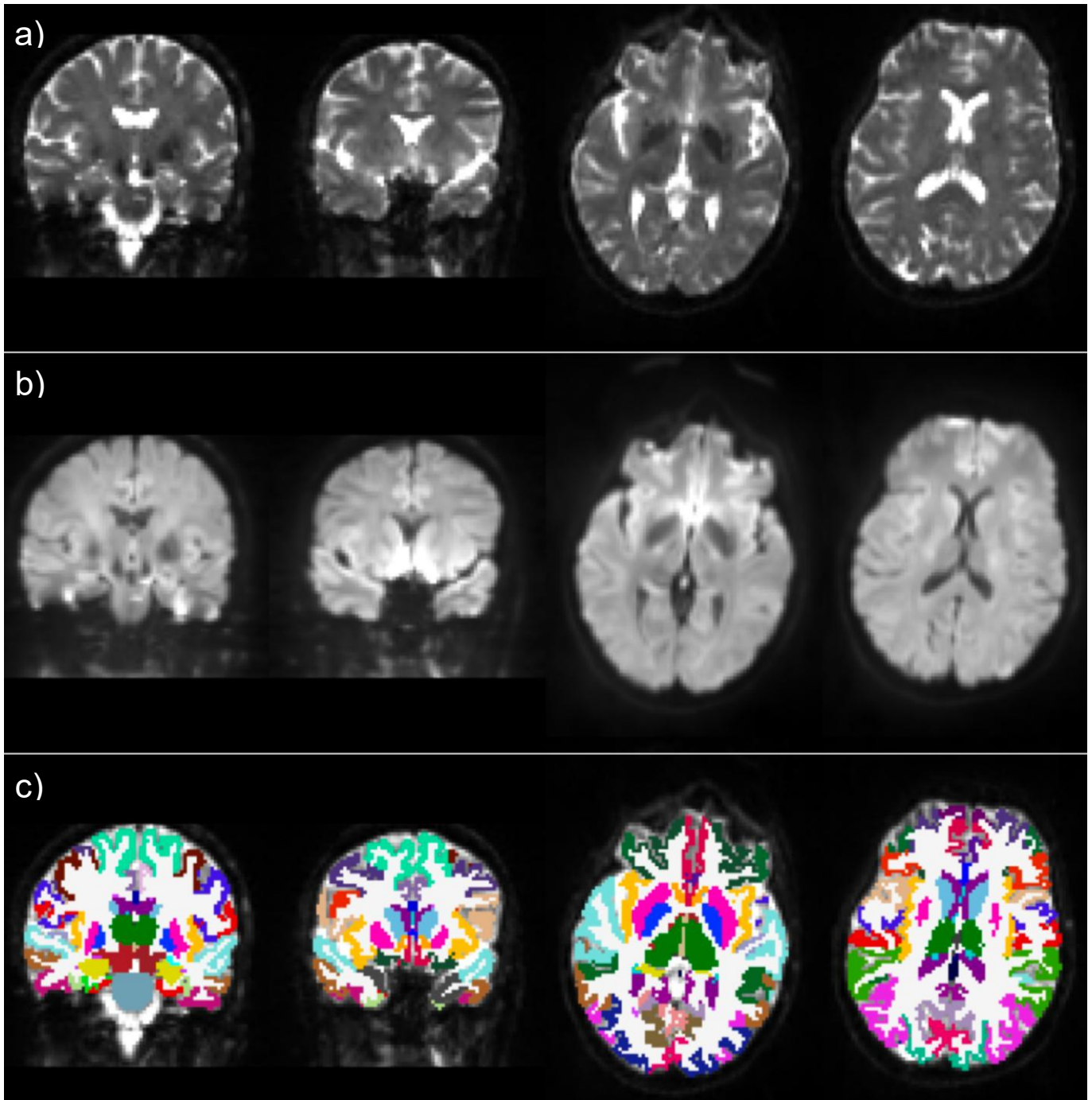
Examples of `dt_recon` results are shown in Figure 5. The results include diffusion-tensor features that can be readily analyzed. The processed dMRI images can, however, be input to other diffusion analysis pipelines and models.

For instance, we used our [CSA-QBI and Hough tractography](#) Matlab toolbox for further analysis, briefly as follows:

- We applied the `reconCSAODF` command to create [CSA ODFs](#) with the spherical harmonic order of 4 and the regularization parameter  $\delta=0.001$ .
- We created a binary mask by loading each subject's `aparc+aseg.nii.gz` and keeping all nonzero labels excluding those that had the term 'ventri', 'Lat-Vent', 'choroid-plexus', or 'CSF' in them.
- We used the `suggestLambda` command to find an optimal value for the tract length prior for each subject, and computed its median across the dataset, which we later used (as a fixed value) for tractography.
- We applied the `HoughTract` command to run [Hough-transform tractography](#) for each subject.
- We used the `connMatrix` command to compute the structural connectivity for each subject.
- We used the `augConnMatrix` command to [augment the connectivity matrix with indirect connections](#).

A limitation of our method is that our diffusion images were resampled twice: once when correcting for susceptibility distortions and again during `dt_recon`'s eddy correction. While FSL's `eddy` can use

the `topup` field when available to avoid the second resampling, this functionality could not have been applied in many of our cases where opposite phase-encoding directions were unavailable.



**Figure 5.** Example results from `dt_recon`. a)  $b=0$  frame. b) averaged diffusion weighted images. c) automatic structural segmentation registered to diffusion space.

### Segmentation Registration Quality Control (QC)

Manual QC was performed on the final diffusion images and registered segmentations for all sessions. To standardize evaluation, automated snapshots were generated from three volumes of each session: a  $b=0$  frame, the average of all diffusion-weighted frames, and the structural segmentation in diffusion space (see Figure 5). Each volume was used to address different metrics of image quality.

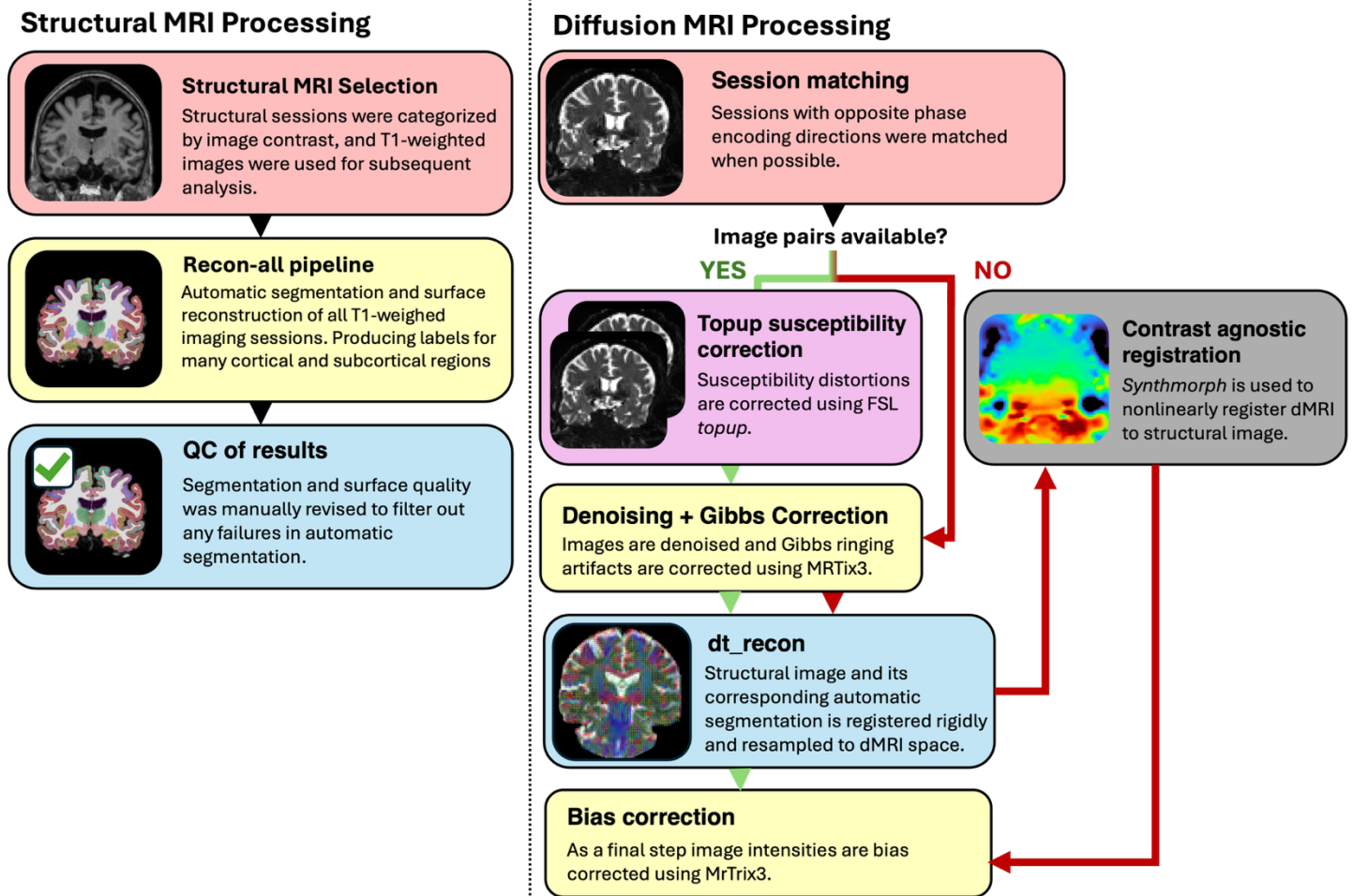
The  $b=0$  volume was used to gauge baseline image quality, field-of-view completeness, bias correction performance, and image contrast. We verified that no relevant anatomy was excluded during image acquisition and inspected overall image contrast. Images in which the  $b=0$  had regions of high intensity relative to the diffusion-weighted image were penalized during QC. The averaged dMRI frames were used to detect motion-correction failures, which appear as ghosting artifacts in the average image. Additionally, overall noise levels and gray-white matter contrast were evaluated. Attention was also given to signal pooling in areas prone to distortion; the pooled signal size and intensity were considered during QC. Although FSL `topup` accounts for signal pooling through intensity modulation, other correction methods do not. Finally, we compared the diffusion image to the registered structural segmentation to identify any misalignments. Although the registration was robust and working well in most cases, particular focus was given to areas prone to distortion artifacts, as these were the most likely to be mislabeled.

A rating system was used to bin the data into five categories of quality considering all the metrics outlined above. From a value of 1, where there were clear failures at some analysis steps (like incorrect segmentation registration, extreme motion or noise, and missing anatomy), to a score of 5, where no artifacts could be seen in the images or analysis. Scores of 3 demonstrated some errors, principally remaining susceptibility errors, which minimally affected segmentation and registration. We included images with a score of 3 or higher in our downstream analysis.

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## Supplemental Figure



**Supplemental Figure.** The flowchart illustrates our processing pipeline for structural MRI and dMRI data from the PPMI dataset. This includes the decision tree for correcting for susceptibility distortions based on the availability of opposite-phase encoding images (green and red arrows).